DATA ANALYSIS PYTHON PROJECT - BLINKIT ANALYSIS

```
! pip install matplotlib seaborn
```

Import Libraries

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
```

import Rawdata

```
df = pd.read_csv(r"C:\Users\Admin\Downloads\drive-download-20261026T064950Z-1-0
```

```
df.head(10)
```

	Item Fat Content	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type	Outlet Size
0	Regular	FDX32	Fruits and Vegetables	2012	OUT049	Tier 1	Medium
1	Low Fat	NCB42	Health and Hygiene	2022	OUT018	Tier 3	Medium
2	Regular	FDR28	Frozen Foods	2010	OUT046	Tier 1	Small
3	Regular	FDL50	Canned	2000	OUT013	Tier 3	High
4	Low Fat	DRI25	Soft Drinks	2015	OUT045	Tier 2	Small
5	low fat	FDS52	Frozen Foods	2020	OUT017	Tier 2	Small
6	Low Fat	NCU05	Health and Hygiene	2011	OUT010	Tier 3	Small
7	Low Fat	NCD30	Household	2015	OUT045	Tier 2	Small
8	Low Fat	FDW20	Fruits and Vegetables	2000	OUT013	Tier 3	High
9	Low Fat	FDX25	Canned	1998	OUT027	Tier 3	Medium

Fileds

```
df.columns
```

```
Index(['Item Fat Content', 'Item Identifier', 'Item Type',
       'Outlet Establishment Year', 'Outlet Identifier',
       'Outlet Location Type', 'Outlet Size', 'Outlet Type', 'Item Visibility',
       'Item Weight', 'Sales', 'Rating'],
      dtype='str')
```

```
df.isnull().sum()
```

```
Item Fat Content              0
Item Identifier               0
Item Type                     0
Outlet Establishment Year     0
Outlet Identifier             0
Outlet Location Type          0
Outlet Size                   0
```

```
Outlet Type                  0
Item Visibility              0
Item Weight               1463
Sales                        0
Rating                       0
dtype: int64
```

```
df.shape
```

```
(8523, 12)
```

```
df.dropna(inplace=True)
```

```
df.shape
```

```
(7060, 12)
```

```
df.columns
```

```
Index(['Item Fat Content', 'Item Identifier', 'Item Type',
       'Outlet Establishment Year', 'Outlet Identifier',
       'Outlet Location Type', 'Outlet Size', 'Outlet Type', 'Item Visibility',
       'Item Weight', 'Sales', 'Rating'],
      dtype='str')
```

Data Cleaning

```
df['Item Fat Content'].unique()
```

```
<StringArray>
['Regular', 'Low Fat', 'low fat', 'LF', 'reg']
Length: 5, dtype: str
```

```
df['Item Fat Content'] = df['Item Fat Content'].replace({'LF':'Low Fat',
                                'low fat':'Low Fat','reg':'Regular'})
```

```
df['Item Fat Content'].unique()
```

```
<StringArray>
['Regular', 'Low Fat']
Length: 2, dtype: str
```

KPI'S

```
df.columns
```

```
Index(['Item Fat Content', 'Item Identifier', 'Item Type',
       'Outlet Establishment Year', 'Outlet Identifier',
       'Outlet Location Type', 'Outlet Size', 'Outlet Type', 'Item Visibility',
       'Item Weight', 'Sales', 'Rating'],
      dtype='str')
```

```
Total_Sales = df['Sales'].sum()
Avg_Sales = df['Sales'].mean()
No_Items = df['Item Type'].count()
Avg_Rating = df['Rating'].mean()

print("Total Sales: ", Total_Sales)
print("Average Sales: ", Avg_Sales)
print("Number of Items: ", No_Items)
print("Average Rating: ", Avg_Rating)
```

```
Total Sales:  997159.2238
Average Sales:  141.24068325779038
Number of Items:  7060
Average Rating:  3.965382436260623
```

Exploratory Data Analysis

```
Total_Sales_by_Fat_Content = df.groupby('Item Fat Content')['Sales'].sum().rese

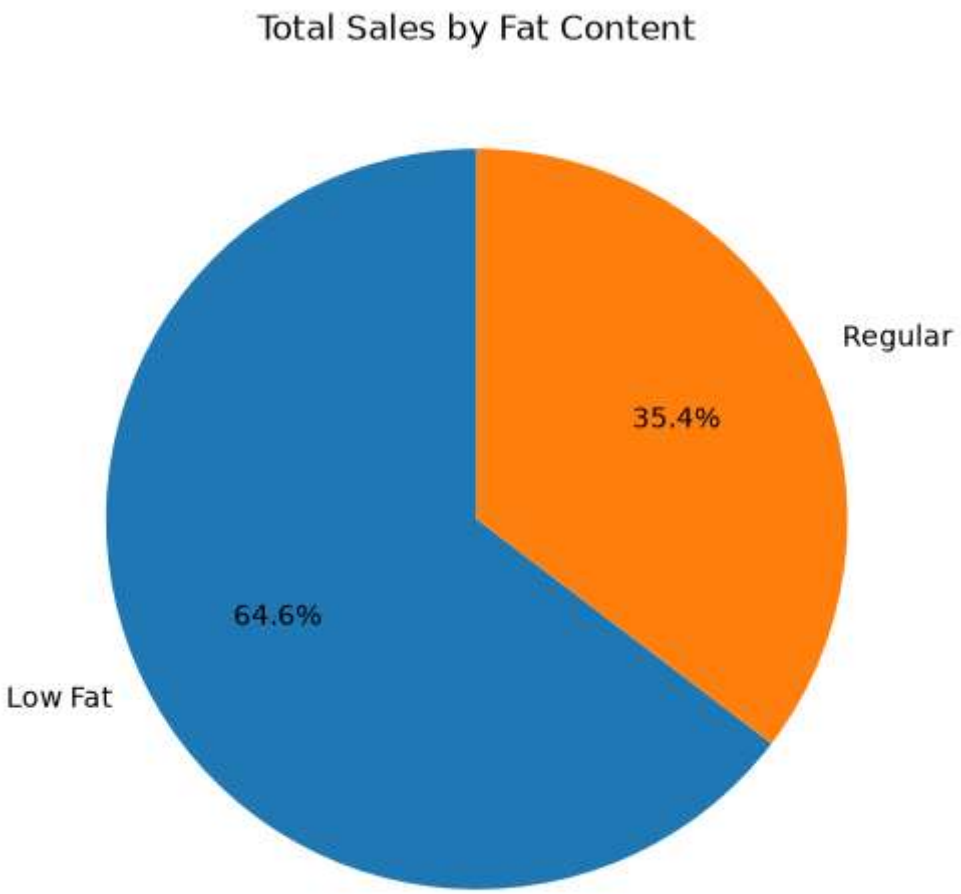
print("Total Sales by Fat Content:", Total_Sales_by_Fat_Content)

# Create Pie Chart
plt.figure(figsize=(6, 6))

plt.pie(
    Total_Sales_by_Fat_Content['Sales'],
    labels=Total_Sales_by_Fat_Content['Item Fat Content'],
    autopct='%1.1f%%',
    startangle=90
)

plt.title('Total Sales by Fat Content')
plt.show()
```

```
Total Sales by Fat Content:   Item Fat Content        Sales
0          Low Fat  644516.7320
1          Regular  352642.4918
```




```
df['Item Type'].value_counts()
```

```
Item Type
Fruits and Vegetables    1019
Snack Foods               988
Household                 759
Frozen Foods              718
Dairy                     566
Canned                    539
Baking Goods              536
Health and Hygiene        430
Soft Drinks               374
Meat                      337
Breads                    204
Hard Drinks               183
Others                    137
Starchy Foods             130
Breakfast                  89
Seafood                    51
Name: count, dtype: int64
```

```
# Calculate total sales by Item Type
Total_Sales_By_Item = df.groupby('Item Type')['Sales'].sum().reset_index()

print(Total_Sales_By_Item)

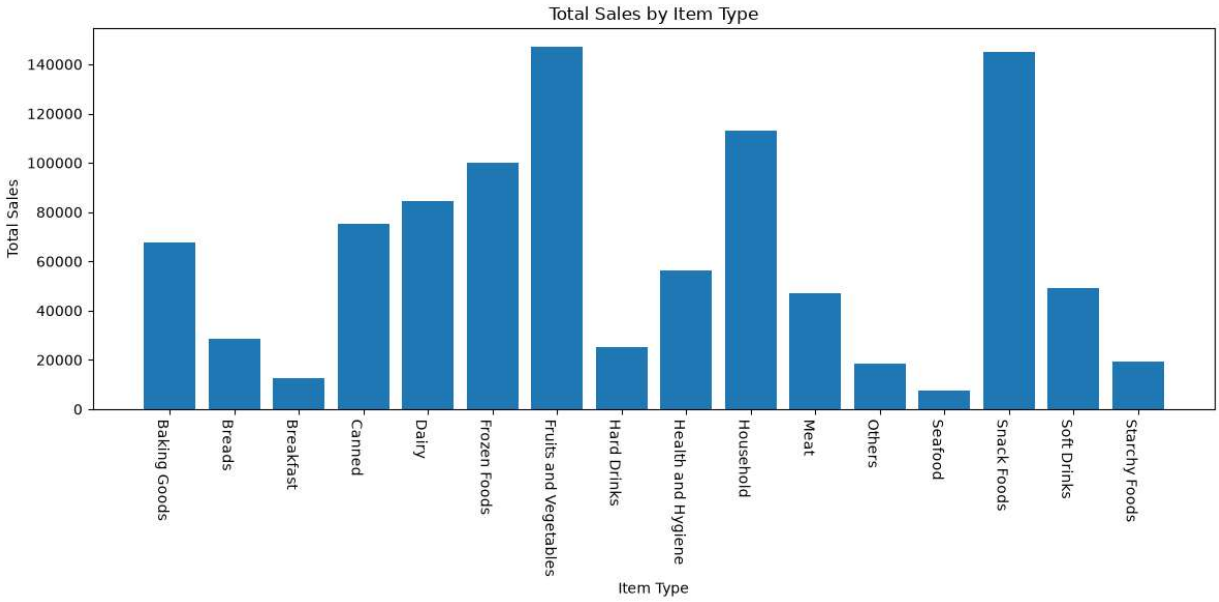
# Create Bar Chart
plt.figure(figsize=(12, 6))

plt.bar(
    Total_Sales_By_Item['Item Type'],
    Total_Sales_By_Item['Sales']
)

plt.title('Total Sales by Item Type')
plt.xlabel('Item Type')
plt.ylabel('Total Sales')

plt.xticks(rotation=-90, ha='right')
plt.tight_layout()
plt.show()
```

```
                Item Type        Sales
0            Baking Goods   67588.1130
1                  Breads   28663.1848
2               Breakfast   12696.1874
3                  Canned   75052.9980
4                   Dairy   84526.4948
5             Frozen Foods   99961.8818
6   Fruits and Vegetables  147188.9632
7              Hard Drinks   25261.6174
8      Health and Hygiene   56383.8020
9                Household  113210.1420
10                   Meat   47159.9010
11                 Others   18624.5970
12                Seafood    7397.5644
13            Snack Foods  144949.1280
14            Soft Drinks   49294.6718
15          Starchy Foods   19199.9772
```



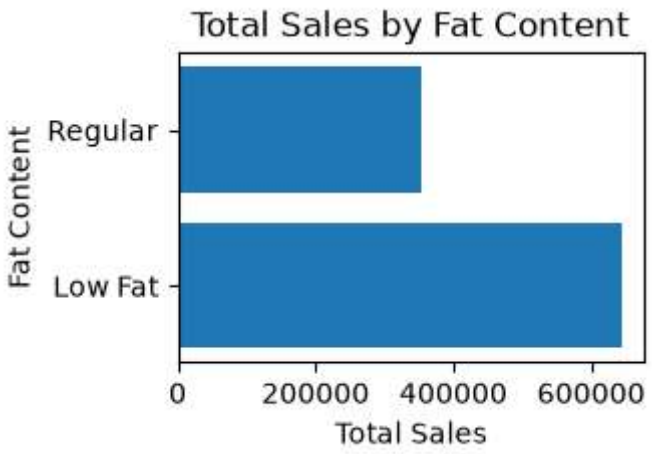

```
Fat_Content_by_Sales = df.groupby('Item Fat Content')['Sales'].sum().reset_inde

print(Fat_Content_by_Sales)

## Create hbar chart
```

```
plt.figure(figsize=(3,2))
plt.barh(
    Fat_Content_by_Sales['Item Fat Content'],
    Fat_Content_by_Sales['Sales']
)
plt.title('Total Sales by Fat Content')
plt.xlabel('Total Sales')
plt.ylabel('Fat Content')
plt.show()
```

```
  Item Fat Content        Sales
0          Low Fat  644516.7320
1          Regular  352642.4918
```




```
Total_Sales_by_Outlet_Establishment = df.groupby('Outlet Establishment Year')['

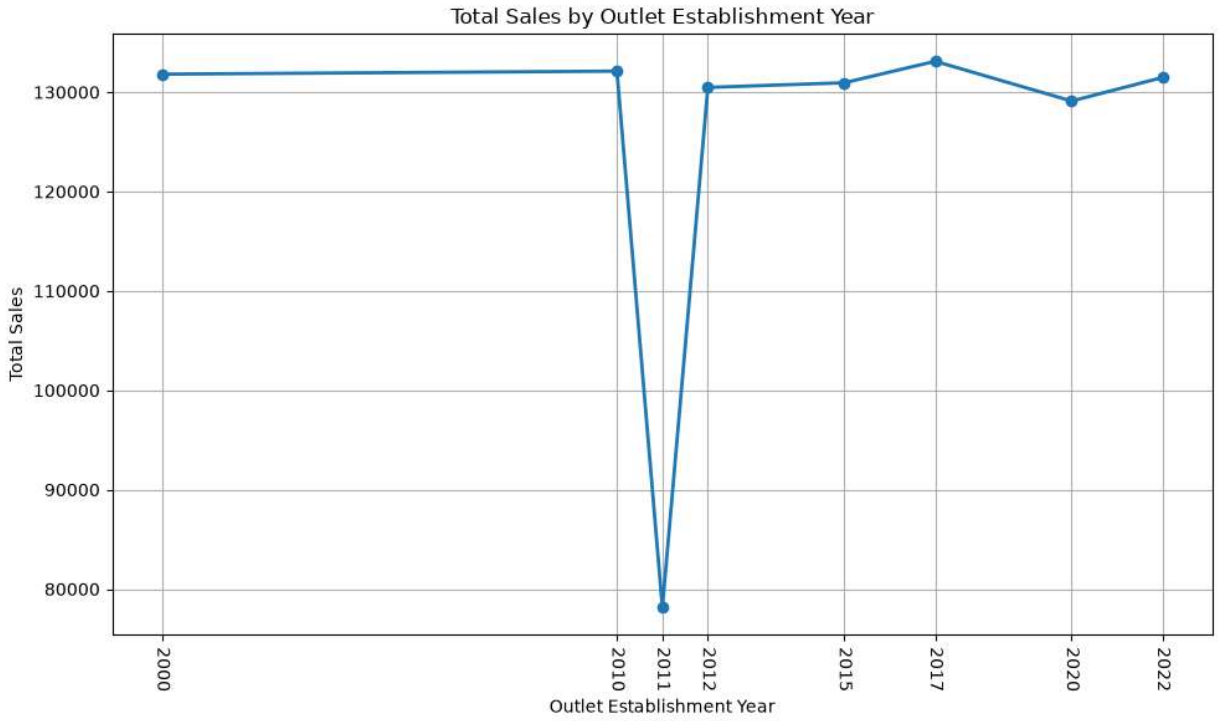
print(Total_Sales_by_Outlet_Establishment)

# Create visualization
plt.figure(figsize=(10, 6))

plt.plot(
    Total_Sales_by_Outlet_Establishment['Outlet Establishment Year'],
    Total_Sales_by_Outlet_Establishment['Sales'],
    marker='o',
    linewidth=2
)

plt.title('Total Sales by Outlet Establishment Year')
plt.xlabel('Outlet Establishment Year')
plt.ylabel('Total Sales')
plt.xticks(
    Total_Sales_by_Outlet_Establishment['Outlet Establishment Year'],
    rotation=-90
)
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
   Outlet Establishment Year        Sales
0                       2000  131809.0156
1                       2010  132113.3698
2                       2011   78131.5646
3                       2012  130476.8598
4                       2015  130942.7782
5                       2017  133103.9070
6                       2020  129103.9564
7                       2022  131477.7724
```




```
Sales_by_Outlet_Size = df.groupby('Outlet Size')['Sales'].sum().reset_index()

print(Sales_by_Outlet_Size)

## create bar chart
plt.figure(figsize=(6, 4))
plt.bar(
    Sales_by_Outlet_Size['Outlet Size'],
    Sales_by_Outlet_Size['Sales']
)
```

```
plt.title('Total Sales by Outlet Size')
plt.xlabel('Outlet Size')
plt.ylabel('Total Sales')
plt.xticks(rotation=-90, ha='right')
plt.tight_layout()
plt.show()
```

```
  Outlet Size        Sales
0        High  248991.5840
1      Medium  377181.0538
2       Small  370986.5860
```

